

WASP-43b

R-0437

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_241221 · created 2026-07-19T13:01:18+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460665.9534 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.017
Transit depth (Rp/Rs)^2	3.92 +/- 0.65 [%]
Orbital Inclination (inc)	80.52 +/- 1.87
Airmass coefficient 1 (a1)	1.013 +/- 0.014
Airmass coefficient 2 (a2)	-0.012 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#3 - [476, 347]
Optimal Method	PSF photometry
Transit Duration (day)	0.047 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.198 ± 0.017 vs 0.1594 (deviation 2.27σ)
Duration fit vs published	0.047 d vs 0.0512 d (deviation 0.51σ)
Mid-time O–C	-5.8 min
Depth SNR	11.6
Residual scatter	1.39 %

Light curve

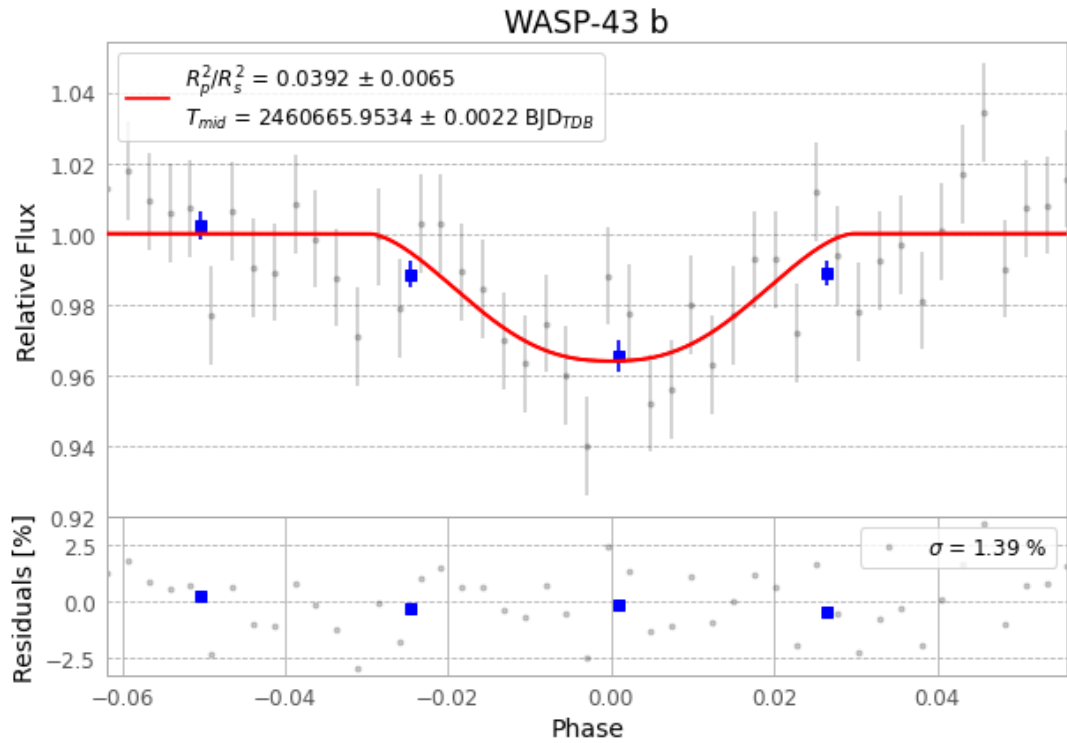


Figure 1 — FinalLightCurve_WASP-43 b_2024-12-21.png (SHA-256 aa9b7c16e1360c86...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [455, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ef8b26a5f84314af...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

47 FITS frames (31 MB), WASP-43241221093615.FITS → WASP-43241221115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-241221.log (eaa17845a5e7...), FinalLightCurve_WASP-43 b_2024-12-21.png (aa9b7c16e136...), FinalLightCurve_WASP-43 b_2024-12-21.csv (bf3beab586af...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 13:01Z.